

Skills Summary

Placing Fractions on a Number Line

Use this reference while completing your worksheet or when studying.

The two counts. On a number line, a fraction is a *distance from 0*, not a picture of a pizza.

Bottom number (denominator): how many equal parts one whole was cut into. Count the *parts*, never the tick marks.

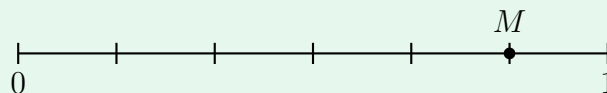
Top number (numerator): how many of those parts you travel, starting at 0.

So $\frac{3}{4}$ means: cut the whole into 4 equal parts, then go 3 of them.

1. Name the point that is already marked

Read the labels at the ends first — they tell you where one whole finishes. Then count the equal parts between 0 and 1 to get the bottom number, and count parts from 0 to the dot to get the top number.

Example: Name point M .



Step 1. The dot is already placed, so the job is to read it — count parts for the bottom number, then count travel for the top number.

Step 2. Equal parts from 0 to 1: six, so the parts are sixths. Parts travelled from 0 to M : five. Five parts of a whole cut into sixths: $M = \frac{5}{6}$

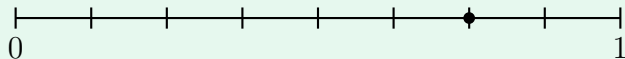
Try it: A line from 0 to 1 is cut into 3 equal parts. A dot sits on the first tick after 0. What fraction names the dot?

check: $\frac{1}{3}$

2. Place a fraction you are given

Work from the bottom number outward: it tells you how many equal parts the whole needs. Then count that many parts from 0 — the top number of them — and put the dot there. Two fractions that land on the same point name the same number, so $\frac{2}{4}$ and $\frac{1}{2}$ are the same place.

Example: Mark $\frac{6}{8}$, then name the point with a simpler fraction.



Step 1. The bottom number 8 says the whole is cut into eighths, so count six eighths from 0 and mark the sixth tick.

Step 2. Now group the eighths in twos: two eighths make one fourth, so the eight parts become four fourths and the dot lands on the third of them. The same point, named in bigger parts: $\frac{3}{4}$

Try it: On a line cut into 6 equal parts, mark $\frac{2}{6}$. What simpler fraction names that same point?

check: $\frac{2}{6}$

3. Compare two fractions on the line

Whichever fraction lands *farther from 0* is the greater one — that is all comparing means on a number line.

$>$ means *is greater than* $<$ means *is less than*

When the top numbers match, the fraction with the *smaller* bottom number wins, because fewer cuts make longer parts.

Example: Which is greater, $\frac{3}{4}$ or $\frac{3}{6}$?

Step 1. Both travel three parts, so the winner is decided by which parts are longer — compare the sizes of the cuts, not the numbers.

Step 2. Cutting a whole into 4 makes longer parts than cutting it into 6, so three of them reach farther. On a line, $\frac{3}{4}$ sits past the middle while $\frac{3}{6}$ sits exactly on it. $\frac{3}{4}$ is farther from 0, so the sign is $>$

Try it: Put $\frac{1}{2}$ and $\frac{5}{8}$ on one eighths line (remember $\frac{1}{2}$ is four eighths). Which sign makes $\frac{1}{2}$ $\frac{5}{8}$ true?

check: $>$

(!) Watch out: count the *spaces* between ticks, not the ticks themselves — a line cut into fourths has five tick marks. And check the label at the right-hand end before you start: if it says 2, the parts between 0 and 1 are still the unit, so a dot past the middle can be more than one whole.